

A Review of Security and Protection Techniques for Machine Learning: Adversarial and Backdoor Perspectives

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Abstract

The rapid expansion of the Internet of Things (IoT) has transformed modern life by connecting physical devices to the internet, creating highly interactive and automated environments. However, this exponential growth has raised severe security concerns. Traditional security mechanisms, such as signature-based Intrusion Detection Systems (IDS), lack the flexibility to handle dynamic and heterogeneous networks, making them vulnerable to zero-day attacks. Machine learning has emerged as a promising paradigm for dealing with evolving threats, yet modern models remain inherently vulnerable to adversarial evasion and backdoor attacks. This paper provides a structured review of threat models, adversarial attacks, backdoor attacks, defense strategies, and evaluation practices. Compared with purely enumerative reviews, it emphasizes a unified classification framework, cross-phase attack-defense relationships, quantitative comparison dimensions, and deployment challenges in IoT and edge intelligence. Recent large-model, multimodal, and adaptive-defense trends are also discussed to improve the timeliness and guiding value of the review.

CCS Concepts

• **Security and privacy** → Intrusion/anomaly detection and malware mitigation; Intrusion detection systems; Security and privacy; Intrusion/anomaly detection and malware mitigation; Malware and its mitigation; Computing methodologies; Machine learning; Machine learning approaches; General and reference; Document types; Surveys and overviews..

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Keywords

adversarial attacks, backdoor attacks, machine learning security, IoT security, edge intelligence

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1 INTRODUCTION

In the context of evolving information technology, the Internet of Things (IoT) has seamlessly integrated everyday objects into the internet, greatly enhancing interactivity and automation. Along with the surge in connected devices, threats to system integrity and availability have become increasingly severe [1]. Intrusion Detection Systems (IDS) play a crucial role in protecting computer systems by monitoring networks in real time to identify anomalous behaviors. However, traditional IDS designed for standard networks face significant bottlenecks in highly heterogeneous IoT environments. Traditional methods rely heavily on known signatures and are often ineffective against advanced threats such as zero-day attacks. To overcome these limitations, machine learning is viewed as a highly promising solution capable of adaptively learning from historical data to identify unknown attacks [2]. Despite the success of deep learning methods in feature extraction, AI security issues have raised substantial academic concern [3]. Attackers can induce misclassifications by applying imperceptible perturbations to input samples or by injecting hidden triggers during training. While numerous studies have examined isolated vulnerabilities, there remains a need for a systematic, cross-phase analysis of these threats in resource-constrained scenarios such as IoT and edge computing. This review therefore makes four contributions: it unifies attack stages, attacker knowledge, goals, capabilities, and constraints into a common taxonomy; compares adversarial and backdoor attacks from lifecycle and deployment perspectives; summarizes defense

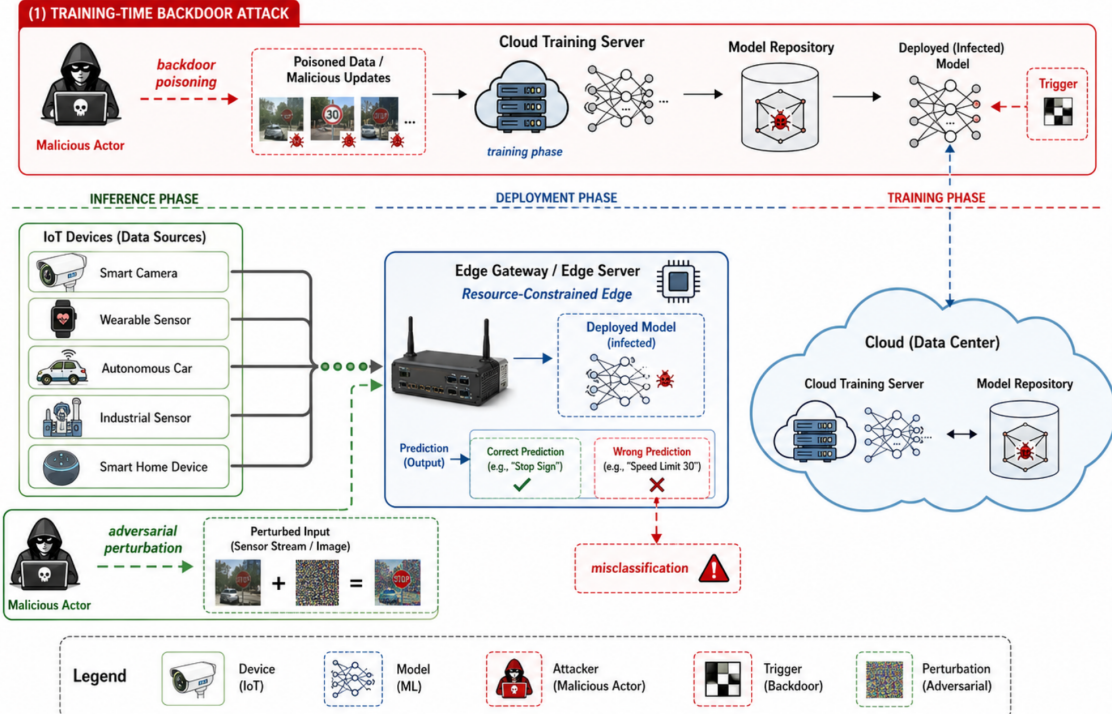


Figure 1: IoT-edge-cloud threat landscape for adversarial and backdoor attacks.

families with quantitative evaluation indicators; and discusses frontier topics such as large-model security, multimodal attacks, and adaptive defense mechanisms for edge intelligence systems.

2 MATERIALS AND METHODS

To ensure a comprehensive and rigorous review of adversarial attacks and defenses in machine learning, particularly in IoT-oriented settings, the relevant literature was systematically surveyed. Eligibility criteria. Peer-reviewed journal articles and high-quality conference papers published predominantly between 2018 and 2026 were prioritized, with special attention to studies from 2022 to 2026 that address large-model security, multimodal attacks, federated edge learning, adaptive defenses, and resource-constrained deployment. Included studies explicitly proposed or evaluated attack or defense mechanisms for machine learning models and reported quantitative outcomes such as Attack Success Rate (ASR), Clean Accuracy (CA), Robust Accuracy (RA), False Positive Rate (FPR), False Negative Rate (FNR), latency, memory consumption, or energy cost. Non-English papers, outdated drafts, and studies without empirical validation were excluded. Data sources and search strategy. Major databases including IEEE Xplore, ACM Digital Library, Scopus, SpringerLink, ScienceDirect, arXiv, and Google Scholar were consulted. Representative search strings combined terms such as Machine Learning, Deep Learning, Large Language Models, Vision-Language Models, Adversarial Attacks, Evasion Attacks, Backdoor Attacks, Poisoning, Defense, Robustness, IoT, Edge Intelligence, Federated Learning, Intrusion Detection, and Multimodal Security.

To ensure that every bibliographic entry is explicitly connected to the manuscript text, this review also draws on early adversarial-example and black-box attack studies, robustness benchmarks, and IDS-security surveys [6, 7, 10–12], representative backdoor, poisoning, and mitigation studies [13–15, 17, 18, 41, 43–46], and general IoT security surveys [16].

3 THREAT MODELS AND ADVERSARIAL EVASION ATTACKS

3.1 Attack Surfaces and Phases

Before analyzing specific attack and defense algorithms, it is essential to establish the security boundaries and formalize a unified threat taxonomy for machine learning models [4]. Attacks targeting the machine learning lifecycle are mainly concentrated in two core phases.

Training phase. Attackers compromise model integrity by injecting malicious samples or tampering with labels in the training set, as in data poisoning and backdoor attacks. In decentralized IoT environments where data is collected from numerous edge sensors, the training pipeline is highly susceptible to such injections [5]. **Inference phase.** After the model is deployed, attackers carefully modify legitimate inputs during testing to force misjudgments, which is particularly dangerous for real-time ML-based IDS, autonomous sensing, and edge perception services where adversarial traffic or sensor streams can bypass detection. Figure 1 connects these two phases with the cloud-edge-IoT deployment stack and highlights

Table 1: Unified threat taxonomy for machine learning attacks.

Dimension	Categories	Description
Attack stage	Training / Inference	When the adversary intervenes
Knowledge access	White-box / Gray-box / Black-box	Attacker’s knowledge of the model
Attack goal	Targeted / Untargeted	Desired attack outcome
Capability	Data modification / Query access / Trigger injection	Operational ability
Constraint	Perturbation budget / Query budget / Physical realizability	Practical limits

Table 2: Comparison of typical evasion attack methods.

Attack Method	Knowledge Required	Attack Type	Computational Cost
FGSM	White-box	Single-step	Low
BIM	White-box	Iterative	Medium
PGD	White-box	Iterative optimization	High
C&W	White-box	Constrained optimization	Very High
AutoAttack	White-box & Black-box	Ensemble benchmark	Extremely High

how poisoned training data, infected repositories, edge deployment, and inference-time perturbations interact in one threat landscape.

3.2 Unified Threat Taxonomy

To systematically evaluate the vulnerabilities of machine learning models, this review organizes the adversary profile along five key dimensions: attack stage, knowledge access, attack goal, attacker capability, and operational constraint. This structure avoids treating attacks as isolated algorithms and instead makes it possible to compare different methods under a common threat-assumption framework.

As shown in Table 1, the unified taxonomy summarizes the major threat-model dimensions used in this review.

3.3 Overview of Evasion Attacks

3.3.1 Basic Principles. The core idea of evasion attacks is to search for a minimal perturbation in the high-dimensional input space such that a small but carefully crafted modification crosses the model decision boundary. Given a clean sample x with label y , the attacker synthesizes an adversarial example $x_{adv} = x + \delta$ under a perturbation constraint, commonly expressed as $\|\delta\|_p \leq \epsilon$, so that the perturbation remains stealthy while still changing the classifier output.

3.3.2 Typical Evasion Attack Methods and Recent Case Studies. Fast Gradient Sign Method (FGSM). FGSM is a single-step gradient-sign attack that uses the gradient of the loss function to generate a perturbation in one step. Recent work in IoT intrusion detection has shown that even a modest perturbation budget can significantly reduce the accuracy of standard ML-based IDS models [8]. Basic Iterative Method (BIM). BIM extends FGSM by applying multiple smaller perturbation steps with clipping after each update. Although slower than FGSM, BIM often produces stronger adversarial examples in localized settings. Projected Gradient Descent (PGD).

PGD is widely regarded as a strong iterative optimization baseline for evaluating adversarial robustness [9]. The comparative characteristics of these evasion attack methods are summarized in Table 2.

4 BACKDOOR ATTACKS

Backdoor attacks can be categorized along multiple dimensions, including trigger form, poisoning strategy, attack scenario, and semantic dependence. This chapter systematically discusses the basic definition and threat model of backdoor attacks, their lifecycle, core taxonomy, representative methods, and comparative analysis with traditional data poisoning and adversarial attacks.^[19,20,40]

4.1 BASIC DEFINITION AND THREAT MODEL

A backdoor attack is commonly regarded as a special form of training-time poisoning and thus belongs to the family of causative attacks against machine learning systems. Its core idea is to implant a malicious behavior pattern during model training so that the infected model preserves normal predictive performance on clean inputs, but outputs an attacker-specified target result whenever a trigger-bearing input is presented at test or deployment time.[19, 20, 34, 40] To characterize this threat in a systematic manner, the threat model is usually described from three perspectives: attacker capabilities, attacker objectives, and attacker constraints. The attacker may partially control the training data, labels, training pipeline, or even a pre-trained model; the objective is to make the model learn a hidden mapping between the trigger and the target label without significantly harming clean accuracy; and the constraints typically concern a low poisoning ratio, strong stealthiness, and feasibility in realistic deployment conditions.

The key difference between backdoor attacks and adversarial attacks lies in the attack stage and the persistence of the attack effect. Adversarial attacks usually operate at inference time and rely on carefully crafted perturbations for the current input, whereas

backdoor attacks are implanted during training and remain latent inside the model. Once the backdoor has been learned, the attacker only needs to present the trigger to reliably induce the target output.^[47,48,49,50]

4.2 ATTACK PROCEDURE AND LIFECYCLE

From a lifecycle perspective, a backdoor attack can be described as a four-stage process: poisoning, training or aggregation, deployment, and triggering. **Poisoning stage.** The attacker embeds backdoor logic through data poisoning, label manipulation, training-pipeline interference, malicious local model updates, or contaminated pre-trained models. A common strategy is to mix a small number of trigger-bearing samples into the training set and relabel them as the attacker-specified target class, forcing the model to learn an abnormal association between the trigger and the target label while minimizing the training loss [19, 20, 34, 39]. **Training or aggregation stage.** In centralized training the poisoned samples are absorbed by the model, whereas in federated learning malicious client updates may be aggregated into the global model. **Deployment stage.** The infected model is distributed to edge gateways, cameras, vehicles, sensors, and other IoT nodes. **Triggering stage.** After deployment, the attacker adds a digital or physical trigger to an otherwise normal input. When the infected model receives a trigger-bearing sample, the abnormal response is activated and the attacker-specified target misclassification is produced. For clean inputs without the trigger, the model usually retains performance close to that of a benign model, which is a major reason why backdoor attacks are difficult to detect through ordinary validation [22, 24–28].

4.3 TAXONOMY OF BACKDOOR ATTACKS

4.3.1 Data-Poisoning Backdoors. Data-poisoning backdoors are the most classical form of backdoor attacks. Their basic strategy is to overlay a fixed trigger on a portion of the training samples and relabel those samples to the attacker-selected target class, so that the model explicitly learns the abnormal mapping “trigger — target label” during training.^[19,20,35] Typical triggers in this category include fixed local patches, visible noise patterns, or blended overlays. Representative methods include BadNets and Blended Attack. Their advantages lie in conceptual simplicity, high attack success rates, and the ability to survive transfer learning in many cases. Their weakness is that the visual content of the poisoned sample often conflicts with its assigned label, which makes such attacks more vulnerable to manual inspection or basic data-cleaning procedures.^[19,20,30,34] In practice, this category mainly threatens third-party data collection, crowdsourced labeling pipelines, and MLaaS outsourcing scenarios where the training data are not thoroughly audited.

4.3.2 Clean-Label Backdoors. Clean-label backdoors implant malicious behavior without changing the original class label of the poisoned sample, thereby significantly improving stealthiness. Unlike traditional data-poisoning backdoors, these poisoned samples remain semantically consistent with their labels and are therefore harder to detect through manual review or automated label-consistency checks.^[21,35,37] Methods in this category typically use optimization in feature space to move target-class samples toward

a source trigger pattern. Hidden Trigger Backdoor Attacks is the canonical representative, while Sleeper Agent further strengthens the effectiveness of this line of attack against models trained from scratch. The main advantage of clean-label backdoors is high concealment; the main drawback is that poisoned-sample construction often requires complex optimization and some prior assumptions about the victim model’s feature representation.^[21,29,37]

4.3.3 Physical Backdoors. Physical backdoors extend the triggering mechanism from the digital domain to the real world. Their triggers are no longer limited to pixel-level patches, but instead appear as physical objects that can be captured by a camera, such as stickers, scarves, eyeglasses, or specially patterned clothing.^[28,38,39,50] Unlike conventional physical adversarial examples, physical backdoors still emphasize training-time contamination: the attacker must make a specific physical object stably correlate with a target class during data collection or model training. Backdoor Attacks Against Deep Learning Systems in the Physical World is a representative study, showing that backdoor attacks can directly threaten cyber-physical systems such as autonomous driving, face recognition, and intelligent surveillance.^[28,38,47,50] The main limitation is that lighting, viewpoint, distance, and occlusion can substantially affect trigger reliability in real environments, so the attack design must also account for physical robustness.

4.4 REPRESENTATIVE METHODS AND PAPER COMPARISON

To trace the technical evolution of backdoor attacks from fixed and explicit triggers to dynamic, invisible, natural, and physical triggers, this section summarizes ten representative papers. Table 3 compresses the comparison into dimensions that are suitable for chapter presentation, including method category, trigger form, typical datasets, core contribution, and main limitation.

4.4.1 Evolutionary Analysis of Representative Methods. From the perspective of methodological evolution, backdoor attacks have broadly undergone four stages. The first stage, represented by BadNets and Targeted Backdoor, established the basic paradigm of training-time implantation and test-time triggering. The second stage, represented by Hidden Trigger and Sleeper Agent, shifted toward clean-label designs in which labels remain correct and poisoned samples appear more natural. The third stage, driven by Dynamic Backdoor, Input-Aware, ISSBA, and WaNet, moved the focus from whether the attack works to how to make it more covert and harder to detect. The fourth stage, represented by Refool and Physical World Backdoor, shows that backdoor attacks have expanded into natural scenes and real physical environments [19–22, 35, 36]. Recent work further extends this trajectory to large language models and multimodal foundation models, where triggers may appear as prompt patterns, instruction-tuning artifacts, visual patches, cross-modal cues, or supply-chain contamination in model checkpoints [40, 42, 62, 63]. Among visual methods, Input-Aware, ISSBA, and WaNet remain especially representative. Input-Aware uses an input-aware generator and cross-trigger testing to increase trigger diversity; ISSBA uses sample-specific invisible perturbations to undermine defenses based on shared trigger statistics; and WaNet

Table 3: Overview of representative backdoor-attack papers.

Paper	Year	Category	Trigger form	Typical datasets	Core contribution	Main limitation
BadNets [19]	2017	Classical poisoning	Fixed small patch	MNIST; traffic signs	Shows that training-time trigger-bearing samples can implant a hidden mapping and that the backdoor may survive transfer learning.	The trigger is fixed and explicit, making it easier to detect through anomaly analysis or pattern inversion.
Targeted Backdoor [20]	2017	Weak-attacker poisoning	Relatively hidden fixed pattern	Traffic signs; ImageNet setting	Shows that only a small number of poisoned samples can still yield a high attack success rate under a weak threat model and discusses physical realizability.	Still mainly relies on a fixed backdoor key and becomes less effective under stronger defenses.
Hidden Trigger [21]	2020	Clean-label	Feature-space hidden trigger	CIFAR-10; ImageNet	Keeps labels correct while enabling target misclassification at test time and marks a key shift from explicit to stealthier backdoors.	Poison construction relies on optimization and is limited against models trained from scratch.
Dynamic Backdoor [22]	2020	Dynamic trigger	Random or conditional triggers	CIFAR-10; CelebA	Breaks the fixed-trigger assumption and shows that dynamic triggers can weaken early defenses such as Neural Cleanse.	As an early dynamic framework, it was later surpassed by stronger input-aware methods.
Input-Aware [24]	2020	Input-aware dynamic	Dynamically generated per input	CIFAR-10; GTSRB	Uses diversity loss and cross-trigger testing to improve trigger non-reusability and is a representative dynamic-backdoor method.	Requires an extra generator and raises both training and deployment cost.
ISSBA [25]	2021	Invisible sample-specific	Steganographic sample-specific triggers	CIFAR-10; ImageNet	Uses deep steganography so different poisoned samples carry different invisible triggers.	Depends heavily on encoding and decoding modules and remains weak in interpretability.
WaNet [26]	2021	Geometric warping	Subtle warping field	CIFAR-10; CelebA	Replaces additive noise with geometric warping to improve visual naturalness and evade pixel-anomaly detection.	Requires careful control of the warping field and mainly targets visual tasks.
Refool [27]	2020	Natural backdoor	Natural reflection phenomenon	Multiple visual datasets	Introduces realistic reflection effects as triggers and makes the trigger more consistent with the imaging process.	Depends on modeling a specific natural phenomenon and is less universal than generic patch-based attacks.
Physical World Backdoor [28]	2021	Physical backdoor	Scarves; eyeglasses; other real objects	Face-related datasets	Systematically verifies the feasibility of backdoor attacks in real physical environments.	Strongly affected by lighting, viewpoint, and distance.
Sleeper Agent [29]	2022	Clean-label extension	Gradient-matching hidden trigger	ImageNet	Uses gradient matching and data selection to improve hidden-trigger attacks against models trained from scratch.	Construction cost is high and the implementation is complex.

Table 4: Multi-dimensional comparison between backdoor attacks and adversarial attacks.

Dimension	Backdoor attacks	Adversarial attacks
Attack stage	Training stage or pre-training / fine-tuning stage	Inference stage or deployment-time testing
Need for training contamination	Usually yes	Usually no
Trigger mechanism	Relies on a preset trigger or hidden pattern	Relies on a perturbation crafted for the current input
Persistence	Persistent and repeatedly activatable	Usually tied to a single sample or a local input instance
Attack objective	Targeted misclassification under the trigger condition	Change the current prediction, either in a targeted or untargeted manner
Effect on clean samples	Usually small; preserving clean performance is important	Generally does not modify model parameters
Defense focus	Data sanitization, model cleaning, abnormal-mapping detection	Adversarial training, input preprocessing, robustness enhancement
Typical threat scenarios	Model-supply-chain contamination, outsourced training, propagation of poisoned pre-trained models	Online-interface attacks, content evasion, physical adversarial examples

employs geometric warping to achieve a high degree of visual naturalness. Together, these methods show that backdoor triggers are evolving from fixed and explicit patterns toward dynamic, invisible, structured, and cross-modal forms [24–26].

4.5 EVALUATION METRICS

Evaluation of backdoor attacks usually considers four aspects simultaneously: attack effectiveness, model utility, robustness, and stealthiness. These dimensions jointly measure whether the attack can be triggered, whether it is difficult to detect, and whether it remains effective under intervention. [40, 42] Attack Success Rate (ASR). ASR measures attack effectiveness. It is the proportion of trigger-bearing test samples whose original labels are not the target label but are nevertheless classified as the attacker’s target label. A higher ASR indicates that the backdoor can be activated more reliably. [19, 20, 24]

$$ASR = N_{\text{trigger} \rightarrow \text{target}} / N_{\text{trigger, total}} \quad (1)$$

Clean Accuracy (CA). CA measures the utility of the infected model on the clean test set. A stealthy backdoor attack usually requires the CA of the infected model to remain close to that of a benign baseline; otherwise, the performance drop would be exposed during ordinary validation.

$$CA = N_{\text{clean, correct}} / N_{\text{clean, total}} \quad (2)$$

4.6 COMPARATIVE ANALYSIS

4.6.1 Backdoor Attacks vs. Adversarial Attacks. Although both kinds of attacks may cause erroneous model outputs, they differ fundamentally in attack timing, required conditions, and threat model. Backdoor attacks implant an abnormal mapping during training and activate it stably during deployment through a trigger, whereas adversarial attacks usually perturb the current input at inference time to exploit the fragility of the decision boundary.[47–49] Table 4 summarizes these differences across attack stage, trigger mechanism, persistence, objective, and defense focus.

4.7 SUMMARY

This chapter has systematically reviewed the basic definition, threat model, lifecycle, taxonomy, representative methods, and evaluation metrics of backdoor attacks. Overall, backdoor attacks have evolved from early fixed and explicit patch triggers to more sophisticated forms such as clean-label, input-aware, sample-specific, geometric-warping, natural-trigger, and physical-world attacks. The technical trajectory shows a sustained shift from static to dynamic triggers, from visible to invisible manipulation, and from the digital domain to realistic deployment settings.[19, 24–29] The core challenge of backdoor attacks lies in their directionality, persistence, and stealthiness. Infected models often maintain high accuracy under routine testing, yet expose stable malicious behavior under specific trigger conditions. As the ecosystem of open pre-trained models, outsourced training services, and model supply chains continues to expand, training-time security risks have become an important issue in trustworthy artificial intelligence.[23, 30–33] Future research will pay increasing attention to more natural triggering mechanisms, more complex cross-modal attacks, and evaluation protocols that better reflect real deployment conditions.

5 DEFENSE STRATEGIES

5.1 Scope and Defense Objectives

Prior sections of this survey established that machine learning models deployed in IoT and edge systems face two structurally distinct threats: evasion attacks that manipulate individual inputs at inference time, and backdoor attacks that embed a persistent failure mode during training or model supply. Although both threats can cause misclassification, they differ in where the compromise occurs, what prior knowledge the attacker requires, and which part of the system must be protected. This distinction shapes the design of defenses: methods effective against one family are often ineffective against the other, and the assumptions made by a given

defense determine where it fits in the system lifecycle. Three evaluation challenges appear across both threat families. First, defense efficacy almost always depends on the attack strength assumed during development and testing; a method that performs well against weak baselines may collapse under stronger adaptive evaluation [12, 49, 53]. Second, almost all defenses carry a cost in terms of clean accuracy, training overhead, inference latency, or model complexity, and these costs scale poorly to resource-constrained edge devices. Third, most published defenses are evaluated on controlled benchmarks under idealized threat models; their behavior in real deployments remains an open question [9].

5.2 Defenses Against Adversarial Evasion Attacks

5.2.1 Adversarial Training. Adversarial training remains the most extensively studied empirical defense against evasion attacks. The foundational idea is to include adversarially perturbed examples in the training data so that the learned decision boundary becomes robust to the perturbation distribution encountered during training. Goodfellow et al. used single-step FGSM perturbations for this purpose and showed a measurable improvement in robustness on simple models [6]. Madry et al. placed adversarial training on a more rigorous footing by framing it as a min-max saddle point problem: the inner maximization finds the worst-case perturbation within an ℓ_∞ ball of radius ϵ , while the outer minimization updates model parameters to minimize loss on that adversarial input [9]. PGD-based adversarial training quickly became the standard empirical baseline in the field because it is both principled and strong against a wide range of first-order attacks. Subsequent work addressed the well-known accuracy-robustness tension inherent in standard adversarial training. Zhang et al. proposed TRADES, which introduces an explicit regularization term that decomposes the adversarial risk into a natural accuracy term and a boundary smoothness penalty, providing a direct knob for trading robustness against clean accuracy [51]. Shafahi et al. reduced the computational burden by reusing gradient computations across perturbation steps, cutting the training overhead by a factor of several compared to full PGD training without large accuracy losses [52]. Despite these improvements, adversarial training carries inherent limits. The robustness achieved against one attack family does not necessarily transfer to attack families with different norms or structural properties. Clean test accuracy typically drops by several percentage points, a cost that may be unacceptable in high-precision tasks.

5.2.2 Input Preprocessing. Preprocessing defenses attempt to neutralize adversarial perturbations before they reach the model. The motivating intuition is that adversarial noise is often concentrated in the high-frequency or high-sensitivity directions of the input, so operations that reduce input complexity may also reduce attack effectiveness. Xu et al. systematically studied this idea by proposing Feature Squeezing, which combines spatial smoothing and color bit-depth reduction to compress the input space, then uses prediction consistency between the original and squeezed versions as an anomaly signal [54]. JPEG compression, median filtering, and randomized resizing have each attracted attention as lightweight options for this preprocessing step. The critical limitation of preprocessing defenses is that they can be defeated by adaptive

attackers who incorporate the preprocessing transformation into their attack objective. Athalye et al. demonstrated this failure mode systematically, showing that methods relying on gradient masking or input randomization often give a false sense of security: once the attacker differentiates through the full pipeline, the claimed robustness frequently disappears [53]. A defense that only hides gradient information rather than altering the decision boundary is not secure against a motivated attacker.

5.2.3 Model Robustification. Model robustification covers a range of techniques that modify the model architecture, loss function, or representation structure to resist adversarial inputs without relying on explicit adversarial data augmentation. Xie et al. proposed feature denoising as an architectural intervention: denoising blocks inserted between layers encourage more stable intermediate representations and showed substantial robustness gains at moderate clean accuracy cost [55]. Jacobian regularization constrains the sensitivity of predictions to small input changes, penalizing large singular values in the input-to-output Jacobian. Several works have observed empirically that wider networks tend to be more robust than narrow counterparts of similar depth, which connects architecture search to robustness without requiring adversarial training directly. The main limitation of robustification techniques is task specificity. The effectiveness of a particular regularization or architectural change is difficult to predict across domains, and combinations of methods do not always compose favorably. In many published comparisons, model robustification provides smaller absolute gains than strong adversarial training while adding engineering complexity.

5.3 Defenses Against Backdoor Attacks

5.3.1 Backdoor Detection. Backdoor detection methods try to determine, before deployment, whether a model or dataset carries a hidden trigger. Neural Cleanse, proposed by Wang et al., reverses the usual attack direction: given a trained model, it optimizes for the smallest perturbation that can cause misclassification into each possible target class and then applies outlier detection over the resulting per-class trigger sizes [17]. A class whose minimum trigger is significantly smaller than the others is flagged as a likely backdoor target. Neural Cleanse is one of the most widely cited detection baselines because its inversion approach generalizes across trigger designs and does not require any poisoned samples from the original training set. Several complementary approaches tackle detection from different directions. Spectral Signatures, proposed by Tran et al., applies singular value decomposition to the feature representations of training samples and exploits the observation that poisoned samples occupy a distinguishable subspace of the representation, detectable as outliers in the principal component projection [30].

5.3.2 Model Inspection. Model inspection methods analyze the internal behavior of a neural network rather than its inputs or outputs. ABS (Artificial Brain Stimulation), proposed by Liu et al., directly stimulates internal neurons by fixing their activation values and observing which activation patterns are sufficient to cause the model to predict a specific class regardless of the input [56]. Neurons that exhibit this behavior are candidates for carrying backdoor

functionality. ABS does not require any clean or poisoned reference data, which is advantageous in audit settings where only the model itself is available. Meta Neural Analysis approaches train a second model to predict whether a given model is trojaned, using features extracted from its behavior on a small reference dataset. While effective when training and reference distributions are aligned, these methods are sensitive to differences in architecture and training regime between the audited model and the meta classifier’s training distribution. In practice, portable model inspection tools must balance coverage of diverse architectures against the cost of collecting training data for the meta model.

5.3.3 Pruning and Fine-Tuning. When a model is suspected to carry a backdoor but retraining from scratch is not available, repair methods based on pruning and fine-tuning offer a practical path. Fine-Pruning, proposed by Liu et al., first identifies neurons that are dormant on clean validation data but activate strongly in the presence of specific stimuli, then removes those neurons and fine-tunes the remaining network on clean data to recover clean task performance [18]. The underlying assumption is that backdoor behavior depends on a relatively small set of dormant neurons that are not needed for normal classification but are reliably triggered by the backdoor pattern. Wu and Wang proposed Adversarial Neuron Pruning (ANP), which identifies vulnerable neurons not by dormancy but by adversarially perturbing neuron weights and identifying those where small perturbations cause large drops in clean accuracy [57].

6 DATASETS AND EVALUATION METRICS

6.1 Benchmark Datasets

6.1.1 Standard Vision Benchmarks. Three vision datasets dominate the current adversarial robustness literature. MNIST [58] is a handwritten digit recognition dataset of 70,000 grayscale 28x28 images. Its simplicity makes it useful for proof-of-concept evaluations and for debugging new method implementations, but conclusions drawn exclusively on MNIST should not be generalized to more complex settings. Models can achieve very high clean accuracy and are not meaningfully challenged by early attack variants at practical perturbation budgets. The dataset has no established connection to IoT or edge sensing applications. CIFAR-10 [59] consists of 60,000 32x32 RGB images in ten classes. It has become the de facto standard for studying the tradeoff between clean accuracy and robust accuracy, in large part because the scale is small enough for extensive experimentation yet large enough to reveal real robustness gaps.

6.1.2 IoT and Cyber-Security Benchmarks. Vision benchmarks are insufficient to evaluate defenses in contexts where machine learning monitors network traffic, detects device anomalies, or processes sensor streams. The following IoT-specific datasets are most commonly referenced in recent security studies. N-BaIoT [60] collects traffic snapshots from nine real IoT devices in both benign and botnet-infected states. Features represent statistical summaries of packet streams rather than raw packets, and the classification task is to distinguish normal device behavior from Mirai and BASHLITE botnet traffic. N-BaIoT is particularly relevant for evaluating ML-based anomaly detectors and for studying adversarial feature manipulation in the network domain, where the perturbation constraint

is not L_p distance but plausibility under real network protocols. Edge-IIoTset [61] is a more recent dataset covering fourteen attack categories across industrial IoT protocols including MQTT, Modbus, and CoAP.

6.2 Evaluation Metrics

Clean accuracy measures the proportion of correctly classified examples in the absence of any attack. In security papers, it is important to distinguish this metric explicitly from attacked performance, because many defenses degrade clean accuracy in exchange for robustness gains. A method that reports only robust accuracy improvements while hiding an unexpectedly large clean accuracy drop is difficult to evaluate fairly. The TRADES framework makes this tradeoff mathematically explicit and includes clean performance as a first-class term in the objective [51].

6.3 Fair Comparison and Reporting Standards

Incomplete reporting is one of the most persistent problems in adversarial ML evaluation. Two papers reporting “robust accuracy” or “ASR” under the same attack name may be measuring against attacks of very different strength, and direct comparison of their numbers is misleading. To support fair comparison, both primary research papers and survey papers should adhere to the following reporting standards. Threat model specification. Every experiment should state whether the evaluation is white-box or black-box, what information the attacker has access to (model parameters, gradients, output logits, hard labels), and whether the defense itself is known to the attacker during attack generation.

6.4 Summary

Meaningful security evaluation in ML for IoT and edge intelligence requires consistent threat model definitions, explicit attack parameter disclosure, paired utility and security metrics, and computational cost reporting suited to the resource constraints of the deployment target. A standardized workflow should include: defining the attacker knowledge and capability assumptions; selecting both vision and IoT-specific datasets; evaluating clean utility, robust accuracy, ASR, FPR, FNR, and computational overhead jointly; testing adaptive attacks when a defense is known to the attacker; and reporting the cloud-edge hardware tier on which training, certification, detection, or inference is performed.

7 DISCUSSION AND FUTURE OUTLOOK

With the deep integration of AI technologies into IoT and other security-critical infrastructures, the security of machine learning models has become a core research challenge. The attack-defense dynamic in this field increasingly resembles a long-term arms race, and a review paper must therefore go beyond listing attack names. A more useful perspective is to link each method to its attacker capability, lifecycle stage, target system, evaluation protocol, and deployment cost. This paper uses that perspective to organize adversarial evasion, backdoor implantation, defenses, datasets, and metrics into a unified framework.

Several future directions deserve particular attention. First, large-model and multimodal security should be incorporated into the same taxonomy as conventional adversarial and backdoor attacks,

because prompt injection, jailbreak optimization, poisoned instruction tuning, backdoored CLIP-style encoders, and cross-modal triggers all expand the attack surface beyond image classification [40, 42, 62, 63]. Second, defense adaptation mechanisms are needed because static defenses are often bypassed by attackers who know the preprocessing, detection, or auditing procedure. Third, IoT and edge scenarios require resource-aware defenses that explicitly report latency, memory, energy, and update overhead instead of only reporting accuracy. Fourth, lifecycle security must be strengthened.

8 CONCLUSIONS

This review has summarized the major security and protection techniques for machine learning from adversarial and backdoor perspectives. By jointly examining threat models, attack mechanisms, defenses, datasets, and evaluation practices, the paper clarifies how vulnerabilities emerge across both training and inference stages. The revised taxonomy highlights the internal relationships among attacker knowledge, attack objective, operational capability, perturbation or trigger constraints, and lifecycle position. The comparative tables and figures further show that no single defense family dominates across all attack types and deployment settings. Robust design for IoT and edge intelligence therefore depends on layered strategies that combine cloud-side training hardening, model auditing, lightweight edge-side filtering, adaptive monitoring, and rigorous evaluation under transparent threat models. Future work should pay special attention to large-model security, multimodal attacks, cross-modal backdoors, and deployment-aware defense adaptation.

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